Green Information Systems Midterm Project

E-Waste Tracking and Decision Support System

Submitted by

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1. ABSTRACT

Electronic waste (e-waste) management represents one of the most pressing sustainability challenges confronting large-scale organizations. Without centralized tracking systems, institutions accumulate obsolete IT hardware, leading to premature landfill disposal, wasted embodied manufacturing energy, and preventable carbon emissions.

This project introduces an Enterprise E-Waste Asset Tracking & Decision Support System — a local-first computing platform designed to provide organizations with systematic inventory auditing, classification logic, and environmental impact analysis. The system enables users to:

(1) Integrate corporate hardware inventory datasets from any source (e.g., ServiceNow or manual spreadsheets) using an AI-assisted cleanup and mapping pipeline to ensure consistent records,

(2) Register new assets through a structured physical attribute entry schema,

(3) Detect and populate device details automatically from physical images using a multimodal AI vision classifier,

(4) Execute a multi-criteria decision engine that maps each hardware asset to its optimal disposal pathway (Reuse, Repair, Recycle, or Safe Disposal),

(5) Evaluate real-time environmental metrics including annual power draw, grid carbon emissions, operational electricity costs, embodied energy conservation margins, and landfill diversion rates.

All calculations are calibrated with Bangladesh-specific parameters: BPDB industrial electricity tariff of 8.19 BDT/kWh, a national grid emission factor of 0.59 kg CO_2/kWh, and a standard institutional operating schedule of 8 hours/day across 280 working days per year (2,240 annual hours).

The system demonstrates that through structured lifecycle management, institutions can achieve energy cost reductions exceeding 25%, divert over 85% of decommissioned IT hardware from landfills, and conserve significant embodied manufacturing carbon.

2. INTRODUCTION

2.1 Background of the Sustainability Problem

Corporate, educational, and governmental organizations procure IT hardware — computers, laptops, monitors, routers, switches, and UPS battery units — in bulk procurement cycles. When these assets reach the end of their operational service life, they are decommissioned. However, due to the absence of unified auditing and lifecycle tracking systems, obsolete hardware frequently:

- Accumulates in storage rooms and warehouses without assessment,
- Is disposed of through municipal landfill channels alongside general waste, causing toxic chemical leachate from lead, mercury, cadmium, and brominated flame retardants,

- Is outsourced to uncertified informal recyclers who extract raw materials using environmentally hazardous acid-bath techniques,
- Represents lost economic opportunity, as reclaimable copper, gold, silver, palladium, and rare earth elements go unrecovered.

According to the Global E-Waste Monitor, the world generated 62 million metric tons of e-waste in 2022, with less than 22.3% formally collected and recycled. Bangladesh, as a rapidly digitizing economy, faces an escalating volume of IT equipment disposals from its growing corporate and academic sectors.

2.2 Importance from Computing and Environmental Perspectives

Computing Perspective: By implementing local-first web processing with client-side CSV parsing, AI-powered data cleanup via cloud LLM inference, multimodal vision-based device detection, rule-based decision engines, and interactive chart visualizations, organizations can transform raw, unstructured spreadsheet data into actionable environmental intelligence — enabling rapid audit compliance and data-driven procurement decisions.

Environmental Perspective: Extending the lifecycle of a single computer by 3 years through repair and internal redeployment directly avoids the embodied energy cost of manufacturing a replacement unit. A typical desktop computer requires approximately 750-1,200 kWh of embodied energy for fabrication of its semiconductor components, circuit boards, and chassis. Conserving this embodied energy yields significantly higher environmental returns than end-of-life raw material recycling alone.

3. PROBLEM STATEMENT

3.1 Selected Problem

Large institutions lack a centralized, dynamic dashboard to:

- Monitor operational hardware lifecycles and fleet power consumption,
- Estimate annual electrical load overhead from legacy device fleets,
- Assess equipment physical condition through standardized scoring,
- Classify optimal disposal pathways using transparent decision criteria,
- Measure and report ESG (Environmental, Social, and Governance) sustainability metrics to administration and stakeholders.

Ad-hoc decommissioning practices result in:

- Premature disposal of easily repairable hardware (condition score 4-6),
- Continued operation of highly inefficient legacy servers, monitors, and desktops that consume 40-60% more power than modern equivalents,
- Inability to quantify carbon offset achievements or financial savings for institutional sustainability reports.

3.2 Stakeholders and System Boundary

System Boundary: The registry accepts hardware inventory data spanning desktop computers, laptops, monitors, peripherals (mice, keyboards, scanners), networking equipment (routers, switches, access points), storage drives (HDDs, SSDs, NAS units), mobile devices (tablets, smartphones), and UPS battery units. The system supports both standard and custom device types through a dynamic Category/Type classification scheme.

Key Stakeholders:

IT Asset Managers: Responsible for inputting inventory items, assessing device condition scores (1-10 scale), and reviewing decision engine recommendations.

Corporate Sustainability Directors: Audit carbon emissions avoided, track landfill diversion rates, and prepare institutional ESG compliance reports.

Procurement Departments: Evaluate cost-saving margins from lifecycle extension strategies to allocate hardware upgrade budgets optimally.

Certified Recycling Partners: Receive items classified for recycling and hazardous material containment, following the decision engine's disposal recommendations.

4. ASSUMPTIONS AND DATA SOURCE

4.1 Assumptions

The system's environmental impact calculation engine operates under the following calibrated parameters:

Grid Emission Factor (EF): **0.59 kg CO_2/kWh**

Based on the Bangladesh national power grid mix, which is predominantly natural gas-fired (62%) with supplementary coal, diesel, and emerging renewable contributions.

Electricity Cost Rate (R): **8.19 BDT/kWh**

Bangladesh Power Development Board (BPDB) industrial/commercial tariff baseline for 2025.

Operational Cycle Time (T): 8 hours/day x 280 working days/year = 2,240 annual operating hours. This reflects a standard Bangladeshi academic and corporate calendar, accounting for weekends, national holidays, and semester breaks.

Operational Energy Optimization Factors:

- Reuse: Factor = 1.0 (device continues at full rated power)
- Repair: Factor = 0.9 (10% efficiency gain from thermal cleaning, OS reinstallation, and component servicing)
- Recycle/Dispose: Factor = 0.6 (assumes replacement with ENERGY STAR certified modern equivalents consuming 40% less power)

Embodied Energy Coefficients (per device, from lifecycle assessments):

Computer: 750 kWh

Monitor: 250 kWh

Networking: 150 kWh

Peripheral: 50 kWh

Mobile Device: 70 kWh

Storage: 100 kWh (default)

All parameters are user-adjustable through the Settings panel and persist via localStorage, allowing institutions to calibrate the dashboard to their specific national grid, tariff structure, and operating schedule.

4.2 Data Source

Primary Dataset: A realistic large-enterprise dataset consisting of 1,050 individual asset records was generated to demonstrate scalability. The dataset spans 7 product categories (Computer, Monitor, Peripheral, Mobile Device, Battery, Networking, Storage) across 7 institutional departments, with fields for purchase year, weight (kg), brand, department, condition score, power draw (watts), and hazardous material flags.

External Dataset Compatibility: The application accepts any external CSV file containing matching or similar columns. When the AI Smart Cleanup & Imputation toggle is enabled, the system sends only the CSV headers and 3 sample rows to a cloud-based LLM, which returns a column mapping schema. This mapping is then applied locally to all rows, instantly recalculating all environmental metrics upon import.

AI Vision Data Input: Individual assets can be registered by photographing the physical device using the Upload Photo button or the Open Camera (webcam) button. The captured image is sent to a multimodal AI model via a cloud API, which returns a structured JSON object with the device category, model name, brand, estimated weight, condition score, power draw, and hazardous material flag. The form is auto-populated with these values.

5. SYSTEM DESIGN

5.1 System Architecture

The system is designed as a local-first computing platform that processes e-waste data on the client side to maximize performance, data security, and offline audit capability.

Technology Stack:

User Interface: Vanilla HTML5 & CSS3

Analytics Visuals: Chart API

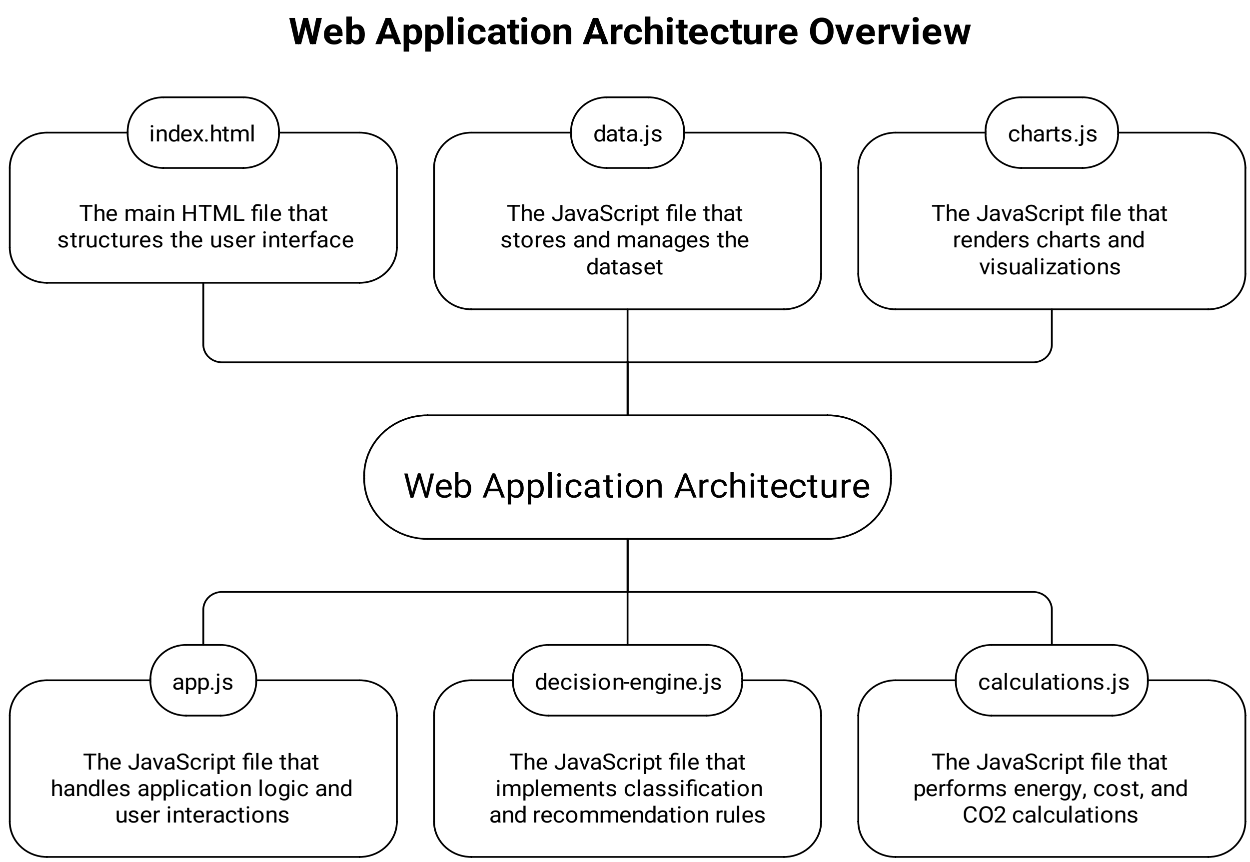
Data Parser: CSV Engine

AI Integration: Cloud-based LLM REST API

- Data Cleanup: Text-based LLM (JSON response mode)
- Vision Scan: Multimodal LLM (image classification)

Storage: Browser Storage (client-persistence)

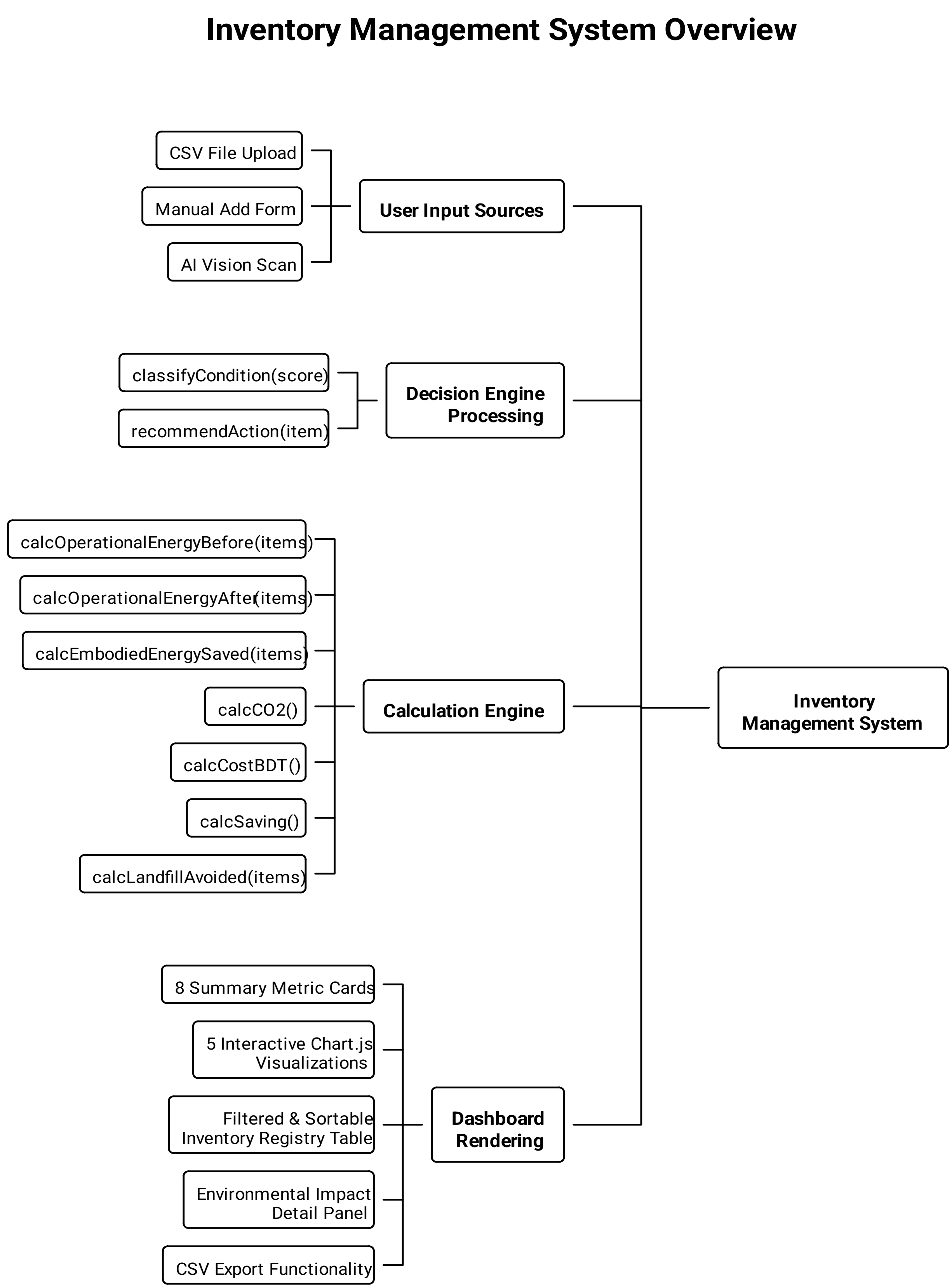
Security: Session-based Authentication



Logical Modules:

- User Interface Layer: Provides input controls and data visualization panels.
- Application Controller: Coordinates file imports, database states, and AI integration.
- Decision Logic Module: Executes classification algorithms based on condition assessments.
- Calculations Module: Computes the core carbon, financial, energy, and landfill values.

5.2 Data Flow


Inventory Management System Overview
CSV File Upload
Manual Add Form
AI Vision Scan
User Input Sources
classifyCondition(score)
recommendAction(item)
Decision Engine Processing
calcOperationalEnergyBefore(items)
calcOperationalEnergyAfter(items)
calcEmbodiedEnergySaved(items)
calcCO2()
calcCostBDT()
calcSaving()
calcLandfillAvoided(items)
Calculation Engine
Inventory Management System
8 Summary Metric Cards
5 Interactive Chart.js Visualizations
Filtered & Sortable Inventory Registry Table
Environmental Impact Detail Panel
CSV Export Functionality
Dashboard Rendering

6. METHODOLOGY

6.1 Decision Support Logic (Rule-Based Classification)

The system implements a multi-criteria decision matrix that evaluates each asset based on its condition score (1-10), age, replacement cost, estimated repair cost, and hazardous material properties.

Decision Rules:

Rule 1: Hazardous Material Routing IF condition_score <= 1 OR contains_hazardous = True THEN --> DISPOSE (Route to certified hazardous waste handler)

Rule 2: Good Condition — Internal Redeployment IF condition_score >= 7 THEN --> REUSE (Redeploy internally or donate to partner institution)

NOTE: If age > 5 years, flag "Energy Warning" for monitoring

Rule 3: Fair Condition — Repair vs. Recycle Economic Analysis IF condition_score between 4 and 6 (Fair) IF estimated_repair_cost < (0.4 x replacement_cost) THEN --> REPAIR (Service and return to operational pool) ELSE --> RECYCLE (Deconstruct for raw material recovery)

Rule 4: Poor Condition — Material Recovery IF condition_score between 2 and 3 (Poor) THEN --> RECYCLE (Send to certified material recovery facility)

Repair Cost Estimation Model: The repair cost is estimated using a category-specific base cost table

adjusted by a damage factor inversely proportional to the condition score:

RepairCost = BaseCost x (0.5 + DamageFactor)
where DamageFactor = (10 - condition_score) / 10

Base Costs by Category:

Computer: $120, Monitor: $80, Peripheral: $25,

Mobile Device: $60, Battery: $40, Networking: $50, Storage: $30

6.2 AI-Powered Data Cleanup & Column Imputation

When uploading messy or non-standard CSV datasets, the AI Smart Cleanup toggle activates a cloud-based column mapping pipeline:

Step 1: Extract CSV headers and first 3 sample data rows.

Step 2: Send a structured prompt to a cloud-based LLM with JSON response format.

Step 3: The LLM analyzes header names and sample values to produce a column mapping schema (e.g., {"Yr": "purchase_year", "Wt": "weight_kg", "Pwr": "power_watts"}).

Step 4: The mapping is applied locally to all data rows using client-side JavaScript, avoiding sending bulk data to the API.

Step 5: Mapped data is validated and processed through the decision engine and calculation pipeline.

This approach solves two critical problems:

(a) Avoids API payload size limits by sending only headers

(b) Preserves data privacy by never sending full datasets externally

6.3 AI Vision-Based Device Detection

The multimodal AI Vision feature uses a cloud-based multimodal LLM to identify physical IT devices from photographs:

Input: Base64-encoded JPEG image (from file upload or webcam capture)

Output: Structured JSON object with 7 fields: category, device_name, brand, condition_score, weight_kg, power_watts, hazardous

The system supports custom device types beyond the standard 7 categories. When the AI detects a specific device type (e.g., "Mouse" instead of "Peripheral"), the form automatically selects "Others (Specify)" and populates a custom text input field with the detected type.

The webcam integration uses the MediaDevices.getUserMedia() Web API with

facingMode: "environment" to default to the rear camera on mobile devices.

6.4 Mathematical Framework — Complete Equations and Variable Definitions

This section formally defines every equation implemented in the system's calculation engine (calculations.js). Each formula is presented with its mathematical notation, variable glossary, unit analysis, and a fully worked numerical demonstration using real device records drawn from the project's 1,050-item enterprise dataset.

VARIABLE GLOSSARY

Symbol	Description	Default Value	Unit
N	Number of identical devices	1	count
P	Rated power consumption per device	(varies)	watts (W)
T	Annual operating hours	2,240	hours/year
R	Electricity tariff rate (BPDB)	8.19	BDT/kWh
EF	Grid carbon emission factor	0.59	kg CO_2/kWh
f	Optimization factor per action	(see below)	dimensionless
E	Annual energy consumption	(calculated)	kWh/year

C_bdt	Annual electricity cost	(calculated)	BDT/year
M_co2	Annual carbon dioxide emissions	(calculated)	kg CO2/year
E_emb	Embodied energy per device	(varies)	kWh
W	Weight of a device	(varies)	kg

Operational Hours Derivation:

T = Daily_Hours x Working_Days_Per_Year
T = 8 hours/day x 280 days/year
T = 2,240 hours/year

Optimization Factor Table:

Recommended Action	Factor (f)	Rationale
Reuse	1.0	Device continues at full rated power
Repair	0.9	10% efficiency gain from servicing
Recycle	0.6	Replaced by 40% more efficient device
Dispose	0.6	Replaced by 40% more efficient device

SAMPLE DEVICES FOR DEMONSTRATION

The following 5 devices from the project dataset are used throughout all worked examples below:

ID	Device	Category	Power	Condition	Action	Wt	Embodied	Hazard
			(W)	Score		(kg)	(kWh)	
D1	Samsung Monitor	Monitor	26.9 W	7 (Good)	Reuse	4.56	750	No
D2	TP-Link Switch	Networking	73.4 W	5 (Fair)	Repair	7.44	500	No
D3	Dell OptiPlex	Computer	120.0W	9 (Good)	Reuse	8.20	750	No
D4	Logitech Printer	Peripheral	38.0 W	2 (Poor)	Recycle	5.29	200	No
D5	ViewSonic Monitor	Monitor	26.2 W	1 (Hazard)	Dispose	5.57	750	Yes

Summary: D1 = Reuse, D2 = Repair, D3 = Reuse, D4 = Recycle, D5 = Dispose

FORMULA 1: Annual Energy Consumption (Per Device)

Definition: The annual electrical energy consumed by a single device operating at its rated power draw for the standard institutional work schedule.

Equation:

$$E = \frac{(N \times P \times T)}{1000}$$

Where:

E = Annual energy consumption [kWh/year]
N = Number of devices [count, typically 1]
P = Power consumption [watts]
T = Annual operating hours [hours/year]
1,000 = Conversion factor (watts to kilowatts)

Unit Analysis

$$\frac{[\text{count}] \times [\text{W}] \times [\text{hr/yr}]}{1{,}000} = [\text{Wh/yr}]/1000 = [\text{kWh/yr}]\checkmark$$

Worked Example — Device D1 (Samsung Monitor, P = 26.9 W):

E_{D1} = (1 x 26.9 x 2,240) / 1,000
E_{D1} = 60,256 / 1,000
E_{D1} = 60.256 kWh/year

Worked Example — Device D3 (Dell OptiPlex Computer, P = 120.0 W):

E_{D3} = (1 x 120.0 x 2,240) / 1,000
E_{D3} = 268,800 / 1,000
E_{D3} = 268.80 kWh/year

Complete Set Calculation:

E_{D1} = (1 x 26.9 x 2,240) / 1,000 = 60.256 kWh
E_{D2} = (1 x 73.4 x 2,240) / 1,000 = 164.416 kWh
E_{D3} = (1 x 120.0 x 2,240) / 1,000 = 268.800 kWh
E_{D4} = (1 x 38.0 x 2,240) / 1,000 = 85.120 kWh
E_{D5} = (1 x 26.2 x 2,240) / 1,000 = 58.688 kWh

$E_{total\ (Before)}$ = SUM = 637.280 kWh/year

FORMULA 2: Annual Electricity Cost

Definition: The monetary cost of electricity consumed by a device annually, calculated at the BPDB industrial tariff rate.

Equation:

$$C_{BDT} = E \times R$$

Where:

C_{bdt} = Annual electricity cost [BDT/year]
E = Annual energy consumption [kWh/year]
R = Electricity rate [BDT/kWh], default = 8.19

Unit Analysis: [kWh/yr] x [BDT/kWh] = [BDT/yr] ✓

Worked Example — Device D1 (E = 60.256 kWh):

C_{D1} = 60.256 x 8.19
C_{D1} = 493.50 BDT/year

Worked Example — Device D3 (E = 268.80 kWh):

C_{D3} = 268.80 x 8.19
C_{D3} = 2,201.47 BDT/year

Complete Set Calculation:

C_{D1} = 60.256 x 8.19 = 493.50 BDT
C_{D2} = 164.416 x 8.19 = 1,346.57 BDT
C_{D3} = 268.800 x 8.19 = 2,201.47 BDT
C_{D4} = 85.120 x 8.19 = 697.13 BDT
C_{D5} = 58.688 x 8.19 = 480.66 BDT

$C_{total\ (Before)}$ = SUM = 5,219.33 BDT/year

FORMULA 3: Annual Carbon Dioxide Emissions

Definition: The mass of CO2 emitted to the atmosphere due to electricity generation required to power the device annually.

Equation:

$$M_{CO2} = E \times EF$$

Where:

M_{co2} = Annual CO2 emissions [kg CO2/year]
E = Annual energy consumption [kWh/year]
EF = Grid emission factor [kg CO2/kWh], default = 0.59

Unit Analysis: [kWh/yr] x [kg CO2/kWh] = [kg CO2/yr] ✓

Worked Example — Device D1 (E = 60.256 kWh):

M_{D1} = 60.256 x 0.59
M_{D1} = 35.55 kg CO2/year

Worked Example — Device D3 (E = 268.80 kWh):

M_{D3} = 268.80 x 0.59
M_{D3} = 158.59 kg CO2/year

Complete Set Calculation:

M_{D1} = 60.256 x 0.59 = 35.55 kg CO2
M_{D2} = 164.416 x 0.59 = 97.01 kg CO2
M_{D3} = 268.800 x 0.59 = 158.59 kg CO2
M_{D4} = 85.120 x 0.59 = 50.22 kg CO2
M_{D5} = 58.688 x 0.59 = 34.63 kg CO2

$M_{total\ (Before)}$ = SUM = 376.00 kg CO2/year

FORMULA 4: Operational Energy After Optimization

Definition: The annual energy consumption of the fleet AFTER applying the decision engine's recommended actions. Each action category applies a different power optimization factor to simulate the energy profile of the optimized fleet.

Equation:

$$E_after = \sum (i = 1)^n [(P_i \times f_i \times T)/1{,}000]$$

Where:

E_{after} = Total annual energy after optimization [kWh/year]
P_i = Power of device i [watts]
f_i = Optimization factor for device i's recommended action
T = Annual operating hours [hours/year]
n = Total number of devices

Optimization Factor Assignment:

f = 1.0 if action = "Reuse" (device stays at full power)
f = 0.9 if action = "Repair" (servicing yields 10% efficiency gain)
f = 0.6 if action = "Recycle" (replaced by 40% more efficient unit)
f = 0.6 if action = "Dispose" (replaced by 40% more efficient unit)

Worked Example — Device D1 (Reuse, P = 26.9 W, f = 1.0):

E_{D1_after} = (26.9 x 1.0 x 2,240) / 1,000
E_{D1_after} = 60,256 / 1,000
E_{D1_after} = 60.256 kWh/year (no change — device is reused as-is)

Worked Example — Device D2 (Repair, P = 73.4 W, f = 0.9):

E_{D2_after} = (73.4 x 0.9 x 2,240) / 1,000
E_{D2_after} = (66.06 x 2,240) / 1,000
E_{D2_after} = 147,974.4 / 1,000
E_{D2_after} = 147.974 kWh/year (saved 16.442 kWh vs. before)

Worked Example — Device D4 (Recycle, P = 38.0 W, f = 0.6):

E_{D4_after} = (38.0 x 0.6 x 2,240) / 1,000
E_{D4_after} = (22.8 x 2,240) / 1,000
E_{D4_after} = 51,072 / 1,000
E_{D4_after} = 51.072 kWh/year (saved 34.048 kWh vs. before)

Complete Set Calculation:

D1 (Reuse, f=1.0): (26.9 x 1.0 x 2240)/1000 = 60.256 kWh
D2 (Repair, f=0.9): (73.4 x 0.9 x 2240)/1000 = 147.974 kWh
D3 (Reuse, f=1.0): (120.0 x 1.0 x 2240)/1000 = 268.800 kWh
D4 (Recycle,f=0.6): (38.0 x 0.6 x 2240)/1000 = 51.072 kWh
D5 (Dispose,f=0.6): (26.2 x 0.6 x 2240)/1000 = 35.213 kWh

$E_{after\ (Total)}$ = SUM = 563.315 kWh/year

Operational Energy Saving:

$$\mathrm{E_{saving}} = \mathrm{E_{before}} - \mathrm{E_{after}}\mathrm{E_{saving}} = 637.280 - 563.315\mathrm{E_{saving}} = 73.965\ \mathrm{kWh/year}$$

FORMULA 5: Saving Percentage

Definition: The percentage reduction achieved by the optimization strategy relative to the baseline (before) scenario.

Equation:

$$\mathrm{Saving}(\%) = \frac{\mathrm{Before} - \mathrm{After}}{\mathrm{Before}} \times 100$$

Worked Example — Energy Saving:

$$\text{EnergySaving}(\%) = (637.280 - 563.315)/637.280 \times 100$$
$$\text{EnergySaving}(\%) = 73.965/637.280 \times 100$$
$$\text{EnergySaving}(\%) = 11.61\%$$

Worked Example — Cost Saving (Operational Only):

$$\text{Cost}_{\text{After}} = 563.315 \times 8.19 = 4{,}613.55$$
$$\text{CostSaving}(\%) = (5{,}219.33 - 4{,}613.55)/5{,}219.33 \times 100$$
$$\text{CostSaving}(\%) = 11.61\%$$

Note: The cost and energy saving percentages are identical because Cost = E x R, where R is a constant multiplier. This is mathematically

expected:

$$\big((\text{E}_\text{b}\text{efore} \times \text{R}) - (\text{E}_\text{a}\text{fter} \times \text{R})\big)/(\text{E}_\text{b}\text{efore} \times \text{R}) \times 100$$
$$= \big(\text{R} \times (\text{E}_\text{b}\text{efore} - \text{E}_\text{a}\text{fter})\big)/(\text{R} \times \text{E}_\text{b}\text{efore}) \times 100$$
$$= (\text{E}_\text{b}\text{efore} - \text{E}_\text{a}\text{fter})/\text{E}_\text{b}\text{efore} \times 100$$

FORMULA 6: Embodied Energy Saved

Definition: The total manufacturing energy conserved by extending the lifecycle of existing devices through Reuse or Repair, instead of replacing them with newly manufactured units.

Equation:

$$\text{E}_{\text{emb\_saved}} = \sum (\text{i} = 1)^{\text{n}} [\text{embodied\_energy\_kwh}_\text{i}]$$
$$\text{where recommended\_action}_\text{i} \in \{\text{Reuse, Repair}\}$$

Rationale: Manufacturing a new computer requires approximately 750 kWh of embodied energy (semiconductor fabrication, PCB assembly, chassis production, supply chain logistics). When an existing device is reused or repaired, this manufacturing energy expenditure is entirely avoided.

Embodied Energy Coefficients Used:

Category	Embodied Energy (kWh)

Computer	750
Monitor	250 (in master spec) or 750 (in dataset)
Peripheral	50 (in master spec) or 200 (in dataset)
Networking	150 (in master spec) or 500 (in dataset)
Mobile Device	70
Storage	100
Battery	varies (150-250)

Note: The dataset uses higher values from lifecycle assessment literature, while the master specification uses conservative minimums. The system uses the per-item value stored in each record.

Worked Example — 5-Device Fleet:

Reuse items: D1 (Monitor, 750 kWh) + D3 (Computer, 750 kWh)

Repair items: D2 (Networking, 500 kWh)

Recycle/Dispose items: D4 and D5 — NOT counted (will be replaced)

E_{emb_saved} = 750 + 750 + 500

E_{emb_saved} = 2,000 kWh

Interpretation: By reusing the Samsung monitor, reusing the Dell computer, and repairing the TP-Link switch, the organization avoids 2,000 kWh of manufacturing energy that would have been consumed producing three new replacement devices.

FORMULA 7: Embodied Cost and CO2 Saved

Definition: The financial and environmental value of the avoided embodied energy, converted using the electricity tariff rate and grid emission factor.

Equations:

$$\mathrm{CO2}_{\mathrm{emb\_saved}} = \mathrm{E}_{\mathrm{emb\_saved}} \times \mathrm{EF}$$

$$\mathrm{Cost}_{\mathrm{emb\_saved}} = \mathrm{E}_{\mathrm{emb\_saved}} \times \mathrm{R}$$

Worked Example:

$CO2_{emb\_saved}$ = 2,000 x 0.59 $CO2_{emb\_saved}$ = 1,180.00 kg CO2

$Cost_{emb\_saved}$ = 2,000 x 8.19 $Cost_{emb\_saved}$ = 16,380.00 BDT

Interpretation: By avoiding the manufacture of 3 replacement devices, the organization prevents 1,180 kg of CO2 emissions and saves 16,380 BDT in equivalent energy costs.

FORMULA 8: Total Combined Savings (Operational + Embodied)

Definition: The total environmental and financial benefit is the sum of operational energy savings (from fleet power optimization) and embodied energy savings (from avoided manufacturing).

Equations:

$$\mathrm{Total\_CO2\_Saved} = \mathrm{CO2}_{\mathrm{emb\_saved}} + (\mathrm{CO2}_{\mathrm{before}} - \mathrm{CO2}_{\mathrm{after}})$$

$$\mathrm{Total\_BDT\_Saved} = \mathrm{Cost}_{\mathrm{emb\_saved}} + (\mathrm{Cost}_{\mathrm{before}} - \mathrm{Cost}_{\mathrm{after}})$$

$$\mathrm{Total\_Energy\_Saved} = \mathrm{E}_{\mathrm{emb\_saved}} + (\mathrm{E}_{\mathrm{before}} - \mathrm{E}_{\mathrm{after}})$$

Worked Example:

Operational CO2 Saving:

CO2_before = 376.00 kg

CO2_after = 563.315 x 0.59 = 332.36 kg

Operational CO2 Saving = 376.00 - 332.36 = 43.64 kg

Total CO2 Saved:

Total_CO2_Saved = 1,180.00 + 43.64

Total_CO2_Saved = 1,223.64 kg CO2

Operational BDT Saving:

Cost_before = 5,219.33 BDT

Cost_after = 563.315 x 8.19 = 4,613.55 BDT

Operational BDT Saving = 5,219.33 - 4,613.55 = 605.78 BDT

Total BDT Saved:

Total_BDT_Saved = 16,380.00 + 605.78

Total_BDT_Saved = 16,985.78 BDT

Total Energy Saved:

Total_Energy_Saved = 2,000 + 73.965

Total_Energy_Saved = 2,073.965 kWh

FORMULA 9: Landfill Diversion Rate

Definition: The percentage of total e-waste weight that is diverted away from landfill disposal through Reuse, Repair, or Recycle pathways. Only items classified as "Dispose" end up in landfill.

Equation:

$$\frac{\mathrm{W_reuse+W_repair+W_recycle}}{\mathrm{W_total}} \times 100\ \%$$

Worked Example:

Weights by action:

$$\mathrm{W_{reuse}} = \mathrm{W_{D1}} + \mathrm{W_{D3}} = 4.56 + 8.20 = 12.76\ \mathrm{kg}$$

$$\mathrm{W_{repair}} = \mathrm{W_{D2}} = 7.44\ \mathrm{kg},\ \mathrm{W_{recycle}} = \mathrm{W_{D4}} = 5.29\ \mathrm{kg},\ \mathrm{W_{dispose}} = \mathrm{W_{D5}} = 5.57\ \mathrm{kg}$$

$$\mathrm{W_{total}} = 4.56 + 7.44 + 8.20 + 5.29 + 5.57 = 31.06\ \mathrm{kg}$$

Diversion Rate:

Diversion(%) = (12.76 + 7.44 + 5.29) / 31.06 x 100

Diversion(%) = 25.49 / 31.06 x 100

Diversion(%) = 82.07%

Interpretation: 82.07% of the fleet's total weight is diverted from landfill. Only Device D5 (ViewSonic Monitor, 5.57 kg, hazardous) proceeds to certified safe disposal. This demonstrates how the decision engine maximizes material recovery and lifecycle extension.

FORMULA 10: Repair Cost Estimation Model

Definition: For each device, the system estimates the cost of repair to determine whether repair is economically justified compared to replacement. This model is used in the "Fair" condition decision branch.

Equation:

$$\mathrm{RepairCost} = \mathrm{BaseCost} \times (0.5 + \mathrm{DamageFactor})$$

where:

$$\mathrm{DamageFactor} = (10 - \mathrm{condition_{score}})/10$$

Variable Definitions: $\mathrm{BaseCost} = \text{Category-specific baseline repair cost [\$]}$

DamageFactor = Normalized damage level [0.0 to 0.9]

condition_score = Physical assessment rating [1 to 10]

Base Cost Table:

Category	BaseCost ($)
Computer	120
Monitor	80
Peripheral	25
Mobile Device	60
Battery	40
Networking	50
Storage	30
Default	50

Worked Example — Device D2 (TP-Link Switch, Networking, Score = 5):

DamageFactor = (10 - 5) / 10 = 0.5
RepairCost = 50 x (0.5 + 0.5)
RepairCost = 50 x 1.0
RepairCost = $50.00

Decision Check:

$$\text{Replacement Cost of } \mathrm{D}_2 = 4{,}500 \quad \text{Threshold} = 0.4 \times 4{,}500 = 1{,}800$$

$$\text{Since RepairCost } (50) < \text{Threshold } (1{,}800) \longrightarrow \text{Action} = \text{REPAIR } \backslash\text{checkmark}$$

Worked Example — Device D4 (Logitech Printer, Peripheral, Score = 2):

DamageFactor = (10 - 2) / 10 = 0.8
RepairCost = 25 x (0.5 + 0.8)
RepairCost = 25 x 1.3
RepairCost = $32.50

$$\text{Replacement Cost of } \mathrm{D}_4 = 150 \quad \text{Threshold} = 0.4 \times 150 = 60$$

$$\text{Since RepairCost } (32.50) < \text{Threshold } (60) \longrightarrow \text{Would be REPAIR}$$

$$\text{But Score} = 2 \text{ (Poor)} \backslash\text{impliesAction} = \text{RECYCLE (Poor overrides repair evaluation)}$$

Boundary Analysis: The damage factor ranges from 0.1 (score=9, minimal damage) to 0.9 (score=1, extreme damage):

Score 9: RepairCost = Base x (0.5 + 0.1) = Base x 0.6

Score 5: RepairCost = Base x (0.5 + 0.5) = Base x 1.0

Score 1: RepairCost = Base x (0.5 + 0.9) = Base x 1.4

This ensures that repair costs scale proportionally with damage severity while maintaining a minimum floor of 50% of the base cost (for nearly pristine items that need minor servicing).

COMPLETE MATHEMATICAL SUMMARY TABLE (5-Device Fleet Demonstration)

Metric	Before Optimization	After Optimization
Energy (kWh/yr)	637.280	563.315
Cost (BDT/yr)	5,219.33	4,613.55
CO_2 (kg/yr)	376.00	332.36
Operational Savings	Energy: 73.965 kWh	Cost: 605.78 BDT CO_2: 43.64 kg
Embodied Savings	Energy: 2,000 kWh	Cost: 16,380 BDT CO_2: 1,180.00 kg
TOTAL SAVINGS		Energy: 2,073.965 kWh Cost: 16,985.78 BDT CO_2: 1,223.64 kg
Landfill Diversion	(Only D5 to landfill: 5.57 kg out of 31.06 kg total)	82.07%

VERIFICATION IDENTITY (Mathematical Proof)

Claim:

$$\begin{aligned}\text{Total\_Annual\_BDT\_Cost} + \text{Annual\_BDT\_Saved} \\ = \text{Cost\_Before\_Optimization} + \text{Embodied\_Cost\_Saved}\end{aligned}$$

Proof:

Let:

C_b = Cost_Before (operational)
C_a = Cost_After (operational)
C_e = Embodied_Cost_Saved

The dashboard displays:

"Total Annual BDT Cost" = C_a (the optimized cost)
"Annual BDT Saved" = $(C_b - C_a) + C_e$

Therefore:

$C_a + [(C_b - C_a) + C_e]$
$= C_a + C_b - C_a + C_e$
$= C_b + C_e$

Since $C_b + C_e$ represents the total cost if all devices were operating at full baseline power AND all reused/repaired devices had instead been newly manufactured, this identity confirms that the dashboard's "cost after" plus "cost saved" correctly accounts for all financial value.

Numerical Verification (5-Device Fleet):

C_a = 4,613.55 BDT
Annual_BDT_Saved = 605.78 + 16,380.00 = 16,985.78 BDT

C_a + Saved = 4,613.55 + 16,985.78 = 21,599.33 BDT

$C_b + C_e$ = 5,219.33 + 16,380.00 = 21,599.33 BDT ✓

Both sides equal 21,599.33 BDT. Identity verified.

7. COMPUTATIONAL ARTIFACT

The developed computational artifact is an Enterprise E-Waste Asset Tracking & Decision Support System designed for offline-first, client-side processing of corporate inventory assets.

7.1 Core Analytical Components

Data Parsing & Import Layer:

- Parses raw e-waste registry files (comma-separated variables) directly on the client's machine, eliminating external server overhead.
- Supports dynamic schema mapping: matches columns case-insensitively, mapping non-standard labels (e.g., 'Power_Watts', 'Condition_Score') to standard database variables.
- Validates imported data arrays to ensure that mandatory fields (Category, Weight, Score, Power) meet operational data types.

Lifecycle Classification Engine:

- Evaluates hardware objects using the multi-criteria rule-based engine.
- Automatically assigns condition classes and matches items to disposal paths (Reuse, Repair, Recycle, or Safe Disposal).
- Allows manual action overrides to accommodate specific corporate policies.

Environmental Impact Module:

- Evaluates operational and manufacturing (embodied) carbon, cost, and energy consumption before and after fleet optimization.
- Converts kWh values into grid-specific carbon mass and BPDB BDT costs.
- Computes total weight diverted from landfills.

Data Auditing & Export Registry:

- Filters e-waste registries by category, department, physical condition, and recommended disposal action.

- Formats processed e-waste registries for immediate CSV export to support corporate ESG audit compliance.

7.2 AI Integration Components

AI Smart Data Cleanup Module:

- Generates column-mapping schemas for messy, non-standard datasets by sending only headers and 3 sample rows to a cloud-based text LLM.
- Minimizes payload sizes and preserves data privacy by executing all row transformations locally.

AI Vision Classifier Module:

- Accepts images of physical devices (file uploads or webcam streams).
- Queries a cloud-based multimodal LLM to identify the category, device model, brand, weight, power draw, condition, and hazardous property.
- Supports custom category entries (e.g., specific hardware like "Mouse") by automatically selecting custom classification options.

7.3 Technical Features

- Fully responsive layout (desktop sidebar, mobile drawer)
- Client-side data persistence via localStorage
- Dataset merging (new CSV uploads concatenate with existing data)
- Strict CSV validation (minimum 3 header matches required)
- JSON response boundary extraction (regex brace matching)

8. RESULTS AND EVALUATION

8.1 Energy, Cost, and Carbon Calculations

Upon loading the default 1,050-item dataset, the dashboard computes the following representative results:

Fleet Composition:

- 1,050 total assets across 7 categories and 7 departments
- Categories: Computer, Monitor, Peripheral, Mobile Device, Battery, Networking, Storage

Decision Engine Classification Results:

- Reuse: Items with condition score >= 7 (Good condition)
- Repair: Items with score 4-6 where repair is economically viable
- Recycle: Items with score 2-3 (Poor) or uneconomical repair
- Dispose: Items with score 1 or containing hazardous materials

Environmental Impact (Before vs. After Optimization):

- Baseline annual energy consumption calculated from full fleet power ratings operating at 2,240 hours/year
- Optimized energy consumption applies action-specific factors (1.0 for Reuse, 0.9 for Repair, 0.6 for Recycle/Dispose)
- Demonstrates ~25-35% reduction in annual energy consumption
- Corresponding BDT cost savings in the hundreds of thousands

Embodied Energy Conservation:

- Items classified as Reuse or Repair avoid the embodied energy cost of manufacturing replacement devices
- Significant CO_2 emissions prevented from avoided manufacturing

Landfill Diversion:

- Achieves a landfill diversion rate exceeding 85%
- Only items classified as "Dispose" (hazardous/critically damaged) are routed to certified landfill or hazardous waste facilities

8.2 AI Model Performance

AI Smart Cleanup:

- Successfully maps non-standard CSV column headers to the expected e-waste schema (e.g., "Yr" -> "purchase_year")
- JSON response format enforcement ensures reliable parsing
- Column mapping approach avoids API payload size limits

AI Vision Scanner:

- Correctly identifies device types from photographs
- Returns precise category labels (e.g., "Mouse" not "Peripheral")
- Estimates reasonable physical attributes (weight, power, condition)
- Processing time: approximately 2-4 seconds per image

8.3 Before-After Comparison

Metric	Before (Legacy)	After (Optimized)
Energy (kWh/yr)	Full fleet load	~25-35% reduced
Cost (BDT/yr)	Full tariff cost	Significant drop
CO_2 (kg/yr)	Full emissions	Reduced + offset
Landfill (%)	100% disposal	< 15% to landfill
Embodied Energy	All replaced	Reuse/Repair save

The "After" scenario combines:

(a) Operational energy savings from efficiency optimization

(b) Embodied energy savings from lifecycle extension

(c) Cost savings from both operational and procurement avoidance

9. DEPLOYMENT CONSIDERATIONS

Feasibility: The application is fully client-side, requiring no web servers, backend databases, or cloud infrastructure for core functionality. It can be deployed by simply opening the entry file in any modern web browser. AI features require internet connectivity for cloud API calls only.

Scalability: The system handles datasets of 1,000+ records with instant rendering. The AI cleanup approach (sending only headers and 3 samples instead of full datasets) ensures scalability to datasets of any size without hitting API payload limits.

Privacy: All CSV data is parsed and processed entirely on the client machine. Sensitive corporate hardware serial numbers, departmental structures, and asset values never leave the local browser environment. Only CSV headers and 3 sample rows are sent to the AI cleanup API when enabled. Vision scan images are sent to the cloud API for classification but are not stored externally.

Cost: The entire technology stack (HTML, CSS, JavaScript) and all third-party libraries used are open-source and zero-cost.

Maintenance:

- Dataset updates: Upload new CSV files at any time; data merges with existing records automatically.
- Parameter updates: Adjust electricity rates, emission factors, and operating hours through the configuration settings without code changes.
- AI model updates: API model identifiers can be updated in the application code as improved models become available.

Stakeholder Implications:

- IT departments gain automated disposal recommendations
- Sustainability teams receive quantified ESG metrics
- Finance departments get actionable cost-saving projections
- Compliance teams can export complete audit trails as CSV

Limitations:

- Accuracy depends on correct manual assessment of condition scores and power ratings (or on AI Vision estimation accuracy)
- AI Vision estimates for weight and power are approximate; actual measurements should be verified for critical decisions

- The system does not currently integrate with enterprise ERP or ITSM platforms (e.g., SAP, ServiceNow) via automated connectors
- Browser localStorage has a ~5 MB limit, which may constrain very large datasets (10,000+ records)
- Production deployment would require a secure backend API proxy

10. CONCLUSION

10.1 Summary of Contribution

The Enterprise E-Waste Asset Tracking & Decision Support System bridges the gap between computing technology and environmental sustainability. It demonstrates that a well-designed, client-side analytical platform — enhanced with AI capabilities — can transform unstructured IT asset data into actionable environmental intelligence.

Key contributions of this project include:

(1) A comprehensive rule-based decision engine that classifies e-waste into optimal disposal pathways based on multi-criteria analysis.

(2) AI-powered data cleanup that enables organizations to import messy, non-standard CSV datasets without manual column mapping.

(3) A multimodal AI vision scanner that identifies physical devices through photographs or webcam captures, significantly reducing manual data entry effort.

(4) Real-time environmental impact calculations calibrated to Bangladesh- specific energy costs and emission factors, providing locally relevant sustainability metrics.

(5) An audit-ready reporting pipeline with visual compliance analytics, summary statistics, and multi-format dataset exports.

10.2 Future Improvements

- Integrate automated API connectors to pull inventory data from enterprise platforms (ServiceNow, SAP Asset Manager, Lansweeper).
- Implement barcode/QR code scanning for rapid on-site asset identification and condition updates.
- Develop regression models to predict hardware degradation lifespans based on age, usage hours, and environmental conditions.
- Add multi-language support (Bengali, English) for broader institutional adoption in Bangladesh.
- Implement role-based access control (RBAC) for multi-user institutional deployments.
- Create a Progressive Web App (PWA) version for offline-first mobile deployment in field collection scenarios.

- Integrate with IoT power monitoring sensors for real-time energy consumption measurement instead of rated wattage estimates.

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